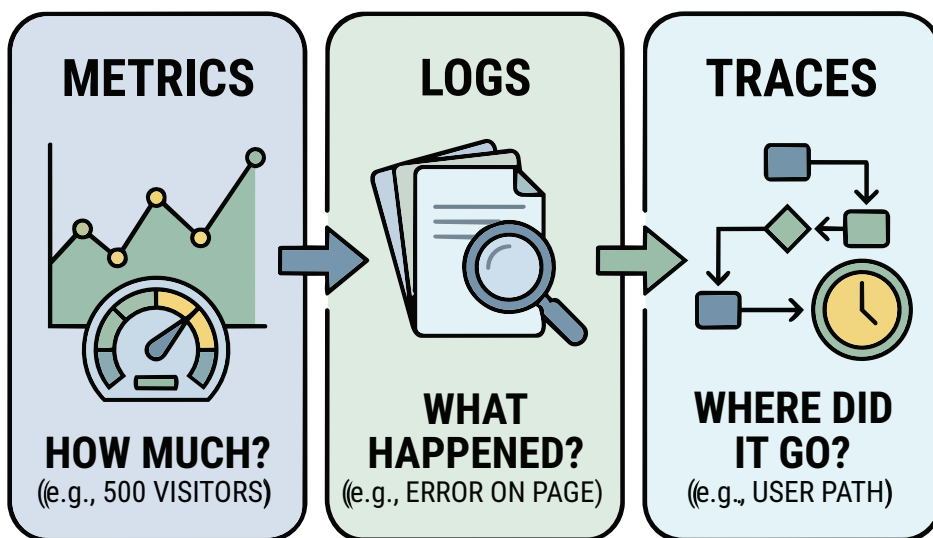


Understanding Metrics, Logs, and Traces without Jargon

The three pillars of observability – metrics, logs and traces – help you find out what went wrong where and why in an AI-based system. You can't do without them.



Three a.m. on a Tuesday and an error shows up. The platform engineer pulls up the logs. The SRE has been watching metrics for the last ten minutes. The developer wants to know why nobody has looked at the trace yet. All three are staring at the same outage from different perspectives.

Most writing on this topic does not help. The articles define spans by saying spans are units of work. They explain cardinality by talking about labels. The reader who came in already lost just stays lost.

Let's try doing things differently by using analogies a non-engineer can follow.

- A *log* records one thing that happened with a timestamp and whatever extra detail the engineer who wrote that code felt like keeping.

- *Metrics* are numbers, sampled over and over, building up a chart you can look at.
- The *trace* is the whole journey of one request, start to finish, across every service it crossed.

Each pillar is built around a different question, and that is genuinely the whole separator.

The phrase 'three pillars of observability' (logs, metrics, traces)

sounds polished, like a settled idea. But it's just shorthand for: Here are three ways to record what is happening inside a running system. Different questions, different recordings. That is the whole insight.

What an event is (and why it matters)

One word sits underneath all three pillars. An event is anything happening

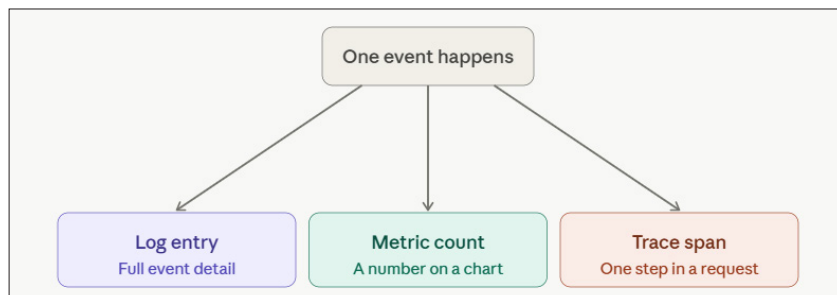


Figure 1: Events and triggers